ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Fusion with X or Y sperm decides the sex.
2	**(d)** 50	1/4 of 200 = 50 (Rryy).
3	**(c)** 1 : 1 : 1 : 1	Four gamete types x one.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Germ-cell DNA is unchanged, so acquired traits are not passed on.
5	**(d)** 1/4	Each birth is independent.
6	**(d)** new combinations of independently inherited genes	Independent assortment gives new combinations.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Masking explains absence of the recessive trait.
8	**(b)** Tt	1 : 1 ratio.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Every F1 plant is Tt; the dominant allele T masks t, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the second statement concerns the sex chromosome carried by sperm and does not explain the equal contribution.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	TT fraction = 1/4 (1); 60 plants (1).	1+1
12	T is dominant over t (1); The tall allele masks the dwarf allele (1).	1+1
13	Ff x ff gives 1 : 1 (1); 2 children (1).	1+1
14	Ratio about 1 : 1 (1); Tall parent Tt, dwarf parent tt (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cross with green wrinkled (yyrr) (1); All yellow round offspring means YYRR (1); Other phenotypes in offspring mean the plant is heterozygous (1).	1+1+1
16	Sperm 23 (22 autosomes) (1); Egg 23 (22 autosomes); zygote 46 (44 autosomes) (1); Liver cell 46 (44 autosomes) (1).	1+1+1
17	A woman has two X chromosomes (XX) (1); Each egg receives only one of them (1); So each egg has one X and 22 autosomes (1).	1+1+1
	OR: Each pregnancy is independent (1); X and Y sperm are equally likely (1); Probability = 1/2 (1).	1+1+1
18	Green beetles are less easily seen (1); They survive and reproduce more (1); The proportion of green beetles increases (1).	1+1+1
	OR: Both parents are heterozygous (1); Each passes the recessive allele (1); The child is homozygous recessive (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Tt (1). (ii) 3 tall : 1 dwarf (1). (iii) $480/4 = 120$ dwarf plants (1). (iv) Genotypic ratio TT : Tt : tt = 1 : 2 : 1 (1); Tt plants $= 480 \times \frac{1}{2} = 240$ (1).	1+1+1+2
20	Tt x tt gives 1 tall : 1 dwarf (1); Rr x Rr gives 3 round : 1 wrinkled (1); Combine the independent ratios (1); Tall round 3, tall wrinkled 1, dwarf round 3, dwarf wrinkled 1 (1); Ratio 3 : 1 : 3 : 1 (1).	1+1+1+1+1
	OR: DNA copying errors (1); Random fusion of gametes (1); Independent assortment of genes (1); Variation helps in adapting (1); Example: people with different skin tones or blood groups (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.
- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1/4	Four equally likely types.
2	**(b)** Mother	Sons get the Y from the father.
3	**(d)** A is false but R is true.	A is false: Tt produces T and t gametes; R is true.
4	**(c)** 9	Nine distinct genotypes.
5	**(d)** 1/4	Both parents are Ff.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Independent assortment.
7	**(b)** 1280	1/16 of total = 80, total = 1280.
8	**(c)** it was ahead of the knowledge of chromosomes and genes	The genetic basis was unknown then.
9	**(c)** A is true but R is false.	Sperms are of two types (X or Y), but all eggs carry only X; R is false.
10	**(d)** A is false but R is true.	The phenotypic ratio is 3 : 1; 1 : 2 : 1 is the genotypic ratio TT : Tt : tt, so A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New types: round green (3) and wrinkled yellow (3) (1); 6/16 = 3/8 (1).	1+1
12	Wrinkled yellow = 3/16 (1); 3/16 x 640 = 120 (1).	1+1
13	All offspring are black, so the black parent has only B (1); Black parent BB, brown parent bb (1).	1+1
14	Each gets one of the mother's two different X chromosomes (1); Autosomes also combine differently in each gamete (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Parents are both Tt (1); Offspring TT, Tt, Tt, tt (1); Tall 3 : dwarf 1 (1).	1+1+1
16	Less hormone, so dwarf plant (1); Giving the hormone makes it taller (1); The offspring still inherit the dwarf allele (1).	1+1+1
17	Homozygous dominant: dominant trait (1); Heterozygous: dominant trait (1); Homozygous recessive: recessive trait (1).	1+1+1
	OR: Test parent is rryy (1); Offspring RrYy, Rryy, rrYy, rryy (1); Round yellow, round green, wrinkled yellow, wrinkled green in 1 : 1 : 1 : 1 (1).	1+1+1
18	He counted a large number of plants in each generation (1); Ratios such as 3 : 1 appeared (1); Ratios led to the idea of dominant and recessive factors (1).	1+1+1
	OR: At least one parent is homozygous dominant (1); The other can be TT or tt (1); Tall allele is dominant (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) XY (1). (ii) X (1). (iii) 50% (1 in 2) (1). (iv) The father determines the sex (1): sperms carry X or Y in equal numbers; an X sperm gives XX (girl) and a Y sperm gives XY (boy), while all eggs carry X (1).	1+1+1+2
20	Each birth is an independent event (1); Probability of a boy is always 1/2 (1); Eggs have X only (1); Sperm may have X or Y (1); Father's sperm decides the sex, not the mother (1).	1+1+1+1+1
	OR: Ratio close to 1 : 1 : 1 : 1 (1); Parent RrYy forms four types of gametes (1); In equal proportion (1); Genes are inherited independently (1); Test parent rryy gives only ry (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.
- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.

- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 46 and 46	Both are diploid.
2	**(a)** Both A and R are true and R is the correct explanation of A.	True explanation.
3	**(d)** 1/16	(1/2)^4.
4	**(d)** there is only one parent and only copying errors	Single parent, little variation.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Gene combination explains it.
6	**(b)** Father	Y comes from the father.
7	**(b)** 1/16	ttrr.
8	**(a)** Both A and R are true and R is the correct explanation of A.	A single gamete type shows F1's gametes.
9	**(a)** Both A and R are true and R is the correct explanation of A.	In Tt the dominant T masks t; the recessive trait shows only in tt.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Rr x rr gives Rr : rr = 1 : 1, so A is true; R is also true but about a different cross and does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	F1 is Tt (1); F2 tt probability is 1/4 (1).	1+1
12	Y sperm gives XY (1); Probability is 0 (the child is a boy) (1).	1+1
13	Each heterozygous gene gives 2 types (1); 2 x 2 x 2 = 8 types (1).	1+1
14	Recessive trait shows only when both alleles are recessive (1); So the genotype has to be homozygous recessive (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Muscle strength gained by exercise is an acquired trait (1); It does not change the DNA of the sperm (1); So it will not be inherited (1).	1+1+1
16	The ratio 3 : 1 is a probability (1); Small numbers may not show the ratio (1); It appears in a large number of offspring (1).	1+1+1
17	Nutrition affects growth (1); Good diet and health can increase height (1); Genes set the potential only (1).	1+1+1
	OR: Each child is a separate fertilisation (1); X and Y sperm are equally likely each time (1); Previous births do not affect the next (1).	1+1+1
18	Dwarf allele was only masked in Tt (1); F2 showed dwarf plants again (1); So it was not lost (1).	1+1+1
	OR: Child is a girl (1); Egg gives an X (1); Zygote is XX (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Transmission of characters/traits from parents to offspring (1). (ii) Small inaccuracies (errors) in DNA copying (1). (iii) DNA of two parents combines, giving new combinations of traits (1). (iv) Some variants are better suited to changing conditions (e.g. heat-resistant bacteria) and survive (1), so the species is not wiped out (1).	1+1+1+2
20	Genes for shape and colour are on different chromosomes (1); Alleles of each gene separate into gametes (1); Four gamete types in equal numbers (1); 16 random combinations (1); 9 : 3 : 3 : 1 ratio results (1).	1+1+1+1+1
	OR: Woman Rr, man rr (1); Gametes R, r and r (1); Children Rr and rr (1); Rr (positive) 1/2 (1); rr (negative) 1/2 (1).	1+1+1+1+1

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